Predictive Analysis Problem Set 6

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2026-03-25

```
rm(list=ls())
library(ISLR)
## Warning: package 'ISLR' was built under R version 4.3.3
head(Hitters)
##                    AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
## -Andy Allanson       293   66     1   30  29    14     1    293    66      1
## -Alan Ashby          315   81     7   24  38    39    14   3449   835     69
## -Alvin Davis         479  130    18   66  72    76     3   1624   457     63
## -Andre Dawson        496  141    20   65  78    37    11   5628  1575    225
## -Andres Galarraga    321   87    10   39  42    30     2    396   101     12
## -Alfredo Griffin     594  169     4   74  51    35    11   4408  1133     19
##                    CRuns CRBI CWalks League Division PutOuts Assists Errors
## -Andy Allanson        30   29     14      A        E     446      33     20
## -Alan Ashby          321  414    375      N        W     632      43     10
## -Alvin Davis         224  266    263      A        W     880      82     14
## -Andre Dawson        828  838    354      N        E     200      11      3
## -Andres Galarraga     48   46     33      N        E     805      40      4
## -Alfredo Griffin     501  336    194      A        W     282     421     25
##                    Salary NewLeague
## -Andy Allanson         NA         A
## -Alan Ashby         475.0         N
## -Alvin Davis        480.0         A
## -Andre Dawson       500.0         N
## -Andres Galarraga    91.5         N
## -Alfredo Griffin    750.0         A
dim(Hitters)
## [1] 322  20
?Hitters
## starting httpd help server ... done
# 1. Data Cleaning
df=na.omit(Hitters) #1. Removing the missing rows
head(df)
##                    AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
```

```
## -Alan Ashby         315   81     7   24  38    39    14   3449   835     69
## -Alvin Davis        479  130    18   66  72    76     3   1624   457     63
## -Andre Dawson       496  141    20   65  78    37    11   5628  1575    225
## -Andres Galarraga   321   87    10   39  42    30     2    396   101     12
## -Alfredo Griffin    594  169     4   74  51    35    11   4408  1133     19
## -Al Newman          185   37     1   23   8    21     2    214    42      1
##                   CRuns CRBI CWalks League Division PutOuts Assists Errors
## -Alan Ashby         321  414    375      N        W     632      43     10
## -Alvin Davis        224  266    263      A        W     880      82     14
## -Andre Dawson       828  838    354      N        E     200      11      3
## -Andres Galarraga    48   46     33      N        E     805      40      4
## -Alfredo Griffin    501  336    194      A        W     282     421     25
## -Al Newman           30    9     24      N        E      76     127      7
##                   Salary NewLeague
## -Alan Ashby        475.0         N
## -Alvin Davis       480.0         A
## -Andre Dawson      500.0         N
## -Andres Galarraga   91.5         N
## -Alfredo Griffin   750.0         A
## -Al Newman          70.0         A
dim(df)
## [1] 263  20
#2. Splitting into Train and Test
set.seed(123)
train.id=sample(1:263,210)
train.data=df[train.id,] #Training data
head(train.data)
##                AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns
## -Marvell Wynne   288   76     7   34  37    15     4   1644   408     16   198
## -Bill Almon      196   43     7   29  27    30    13   3231   825     36   376
## -Rick Cerone     216   56     4   22  18    15    12   2796   665     43   266
## -Jack Howell     151   41     4   26  21    19     2    288    68      9    45
## -Spike Owen      528  122     1   67  45    51     4   1716   403     12   211
## -Tom Paciorek    213   61     4   17  22     3    17   4061  1145     83   488
##                CRBI CWalks League Division PutOuts Assists Errors Salary
## -Marvell Wynne  120    113      N        W     203       3      3    240
## -Bill Almon     290    238      N        E      80      45      8    240
## -Rick Cerone    304    198      A        E     391      44      4    250
## -Jack Howell     39     35      A        W      28      56      2     95
```

```
## -Spike Owen     146    155      A        W     209     372     17    350
## -Tom Paciorek   491    244      A        W     178      45      4    235
##                NewLeague
## -Marvell Wynne         N
## -Bill Almon            N
## -Rick Cerone           A
## -Jack Howell           A
## -Spike Owen            A
## -Tom Paciorek          A
test.data=df[-train.id,] #Test data
head(test.data)
##                AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns
## -Alvin Davis     479  130    18   66  72    76     3   1624   457     63   224
## -Andre Dawson    496  141    20   65  78    37    11   5628  1575    225   828
## -Andy VanSlyke   418  113    13   48  61    47     4   1512   392     41   205
## -Buddy Bell      568  158    20   89  75    73    15   8068  2273    177  1045
## -Barry Bonds     413   92    16   72  48    65     1    413    92     16    72
## -Bobby Bonilla   426  109     3   55  43    62     1    426   109      3    55
##                CRBI CWalks League Division PutOuts Assists Errors Salary
## -Alvin Davis    266    263      A        W     880      82     14    480
## -Andre Dawson   838    354      N        E     200      11      3    500
## -Andy VanSlyke  204    203      N        E     211      11      7    550
## -Buddy Bell     993    732      N        W     105     290     10    775
## -Barry Bonds     48     65      N        E     280       9      5    100
## -Bobby Bonilla   43     62      A        W     361      22      2    115
##                NewLeague
## -Alvin Davis           A
## -Andre Dawson          N
## -Andy VanSlyke         N
## -Buddy Bell            N
## -Barry Bonds           N
## -Bobby Bonilla         N
dim(train.data);dim(test.data)
## [1] 210  20
## [1] 53 20
#3. Best subset selection
library(leaps)
## Warning: package 'leaps' was built under R version 4.3.3
best.subset=regsubsets(Salary~.,data=train.data,nvmax=19)
```

```
summary(best.subset)
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = train.data, nvmax = 19)
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: exhaustive
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns CRBI
## 1  ( 1 )  " "   " "  " "   " "  " " " "  " "   " "    " "   " "    "*"   " "
## 2  ( 1 )  " "   " "  " "   " "  " " "*"  " "   " "    " "   " "    " "   "*"
## 3  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   " "    " "   " "
## 4  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 5  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 6  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 7  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 8  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 9  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 10  ( 1 ) "*"   "*"  " "   " "  " " "*"  " "   "*"    " "   " "    "*"   "*"
## 11  ( 1 ) "*"   "*"  " "   " "  " " "*"  " "   "*"    " "   " "    "*"   "*"
## 12  ( 1 ) "*"   "*"  " "   " "  " " "*"  " "   "*"    " "   " "    "*"   "*"
```

```
## 13  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    " "   " "    "*"   "*"
## 14  ( 1 ) "*"   "*"  " "   "*"  " " "*"   "*"   "*"    " "   " "    "*"   "*"
## 15  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 16  ( 1 ) "*"   "*"  " "   "*"  " " "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 17  ( 1 ) "*"   "*"  "*"   " "  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 18  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 19  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
##           CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 2  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 3  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 4  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 5  ( 1 )  " "    " "     "*"       " "     " "     " "    " "
## 6  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 7  ( 1 )  " "    " "     "*"       "*"     "*"     " "    " "
## 8  ( 1 )  " "    " "     "*"       "*"     "*"     "*"    " "
## 9  ( 1 )  " "    " "     "*"       "*"     "*"     "*"    "*"
## 10  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    " "
## 11  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    "*"
## 12  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 13  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 14  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 15  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 16  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 17  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 18  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 19  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
#Adjusted R squared*/*
adjr2=summary(best.subset)$adjr2
adjr2
##  [1] 0.3916432 0.4742947 0.5077294 0.5327750 0.5492949 0.5541373 0.5573872
##  [8] 0.5609618 0.5631858 0.5668744 0.5702673 0.5721978 0.5720259 0.5710908
## [15] 0.5696920 0.5678958 0.5659643 0.5638778 0.5615944
cp=summary(best.subset)$cp
cp
##  [1] 82.632753 44.219902 29.310338 18.476068 11.723250 10.452933  9.938504
##  [8]  9.290045  9.274000  8.603345  8.083007  8.235300  9.336337 10.776077
## [15] 12.416727 14.225938 16.086206 18.005194 20.000000
bic=summary(best.subset)$bic
```

```
bic
##  [1]  -94.68167 -121.01098 -130.48038 -137.12086 -140.36017 -138.31349
##  [7] -135.53972 -132.93765 -129.70443 -127.19075 -124.55313 -121.21484
## [13] -116.85207 -112.12077 -107.16960 -102.03301  -96.84021  -91.58262
## [19]  -86.24126
coef(best.subset, which.min(bic)) #Best BIC Model
##  (Intercept)        Walks       CAtBat        CHits       CHmRun    DivisionW
##  149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -129.0165012
coef(best.subset, which.min(cp)) #Best cp Model
##  (Intercept)        AtBat         Hits        Walks       CAtBat        CRuns
##  134.1593585   -1.9328699    5.8271814    6.5037245   -0.2063323    1.7820180
##         CRBI       CWalks    DivisionW      PutOuts      Assists   NewLeagueN
##    1.1120229   -0.8879140 -121.2135304    0.1925628    0.3316973   71.1433129
coef(best.subset, which.max(adjr2)) #Best Adjusted R squared Model
##  (Intercept)        AtBat         Hits        Walks       CAtBat        CRuns
##  154.1594365   -1.8722264    5.7547106    6.3852096   -0.2092154    1.7712871
##         CRBI       CWalks    DivisionW      PutOuts      Assists       Errors
##    1.1419024   -0.8989348 -121.8633686    0.2013298    0.5363804   -6.4726568
##   NewLeagueN
##   75.8378982
```

We fit models with 1 to 19 predictors. Then we choose best model using: Adjusted R^2 → maximum BIC → minimum Cp → minimum

```
# 4. Train MSE and Test MSE
m.cp=lm(Salary~AtBat+Hits+Walks+CAtBat+CRuns+CRBI+CWalks+Division+PutOuts+Assists+N
ewLeague,data=train.data)
m.bic=lm(Salary~Walks+CAtBat+CHits+CHmRun+Division,data=train.data)
m.adjr=lm(Salary~AtBat+Hits+Walks+CAtBat+CRuns+CRBI+CWalks+Division+PutOuts+Assists
+Errors+NewLeague,data=train.data)
mse.train.cp=1/210*(sum((train.data$Salary-predict(m.cp))^2)) #Training MSE for cp
mse.train.bic=1/210*(sum((train.data$Salary-predict(m.bic))^2)) #Training MSE for b
ic
mse.train.adjr=1/210*(sum((train.data$Salary-predict(m.adjr))^2)) #Training MSE for
adjR2
mse.train.bic
## [1] 97026.43
mse.train.adjr
## [1] 88935.8
mse.train.cp
## [1] 89790.63
y.hat.test.cp=predict(m.cp, newdata=test.data)
```

```
y.hat.test.bic=predict(m.bic, newdata=test.data)

y.hat.test.adjr=predict(m.adjr, newdata=test.data)

mse.test.cp=mean((test.data$Salary-y.hat.test.cp)^2)

mse.test.cp

## [1] 125652.3

mse.test.bic=mean((test.data$Salary-y.hat.test.bic)^2)

mse.test.bic

## [1] 153090.5

mse.test.adjr=mean((test.data$Salary-y.hat.test.adjr)^2)

mse.test.adjr

## [1] 130066.9

out.best=data.frame(Measure=c("Cp","BIC","AdjR2"),Training_MSE=c(mse.train.cp,mse.t
rain.bic,mse.train.adjr),

                    Test_MSE=c(mse.test.cp,mse.test.bic,mse.test.adjr))

out.best

##   Measure Training_MSE Test_MSE

## 1      Cp     89790.63 125652.3

## 2     BIC     97026.43 153090.5

## 3   AdjR2     88935.80 130066.9
```

In all cases we observe:

Training MSE < Test MSE This is expected because model is trained on training data

Observations on Model Comparison:

Model with lowest Training MSE → Best fit on training data Model with lowest Test MSE → Best for prediction (preferred)

```
# 5. Forward Selection

forward.subset=regsubsets(Salary~.,data=train.data,nvmax=19,method="forward")

summary(forward.subset)

## Subset selection object

## Call: regsubsets.formula(Salary ~ ., data = train.data, nvmax = 19,

##     method = "forward")

## 19 Variables  (and intercept)

##            Forced in Forced out

## AtBat          FALSE      FALSE

## Hits           FALSE      FALSE

## HmRun          FALSE      FALSE

## Runs           FALSE      FALSE

## RBI            FALSE      FALSE

## Walks          FALSE      FALSE

## Years          FALSE      FALSE
```

```
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: forward
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns CRBI
## 1  ( 1 )  " "   " "  " "   " "  " " " "  " "   " "    " "   " "    "*"   " "
## 2  ( 1 )  " "   " "  " "   " "  " " "*"  " "   " "    " "   " "    "*"   " "
## 3  ( 1 )  " "   " "  " "   " "  " " "*"  " "   " "    " "   " "    "*"   " "
## 4  ( 1 )  " "   " "  " "   " "  " " "*"  " "   " "    " "   " "    "*"   " "
## 5  ( 1 )  " "   " "  " "   " "  " " "*"  " "   " "    " "   " "    "*"   "*"
## 6  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    " "   " "    "*"   "*"
## 7  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   " "    "*"   "*"
## 8  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   " "    "*"   "*"
## 9  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   " "    "*"   "*"
## 10  ( 1 ) " "   " "  " "   " "  " " "*"  " "   "*"    "*"   " "    "*"   "*"
## 11  ( 1 ) " "   " "  " "   " "  " " "*"  " "   "*"    "*"   " "    "*"   "*"
## 12  ( 1 ) " "   " "  " "   " "  " " "*"  "*"   "*"    "*"   " "    "*"   "*"
## 13  ( 1 ) "*"   " "  " "   " "  " " "*"  "*"   "*"    "*"   " "    "*"   "*"
## 14  ( 1 ) "*"   "*"  " "   " "  " " "*"  "*"   "*"    "*"   " "    "*"   "*"
## 15  ( 1 ) "*"   "*"  " "   " "  " " "*"  "*"   "*"    "*"   "*"    "*"   "*"
## 16  ( 1 ) "*"   "*"  " "   "*"  " " "*"  "*"   "*"    "*"   "*"    "*"   "*"
## 17  ( 1 ) "*"   "*"  " "   "*"  "*" "*"  "*"   "*"    "*"   "*"    "*"   "*"
## 18  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"  "*"   "*"    "*"   "*"    "*"   "*"
## 19  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"  "*"   "*"    "*"   "*"    "*"   "*"
##           CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 2  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 3  ( 1 )  "*"    " "     " "       " "     " "     " "    " "
## 4  ( 1 )  "*"    " "     "*"       " "     " "     " "    " "
```

```
## 5  ( 1 )  "*"    " "     "*"       " "     " "     " "    " "
## 6  ( 1 )  "*"    " "     "*"       " "     " "     " "    " "
## 7  ( 1 )  "*"    " "     "*"       " "     " "     " "    " "
## 8  ( 1 )  "*"    " "     "*"       "*"     " "     " "    " "
## 9  ( 1 )  "*"    " "     "*"       "*"     " "     " "    "*"
## 10  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    "*"
## 11  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 12  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 13  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 14  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 15  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 16  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 17  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 18  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 19  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
coef(forward.subset, which.min(summary(forward.subset)$bic)) #Best BIC Model
##  (Intercept)        Walks       CAtBat        CHits        CRuns         CRBI
##  127.8159745    6.4746441   -0.4127407    1.1133040    1.0341530    0.8520825
##       CWalks    DivisionW
##   -0.5432536 -127.3877839
coef(forward.subset, which.min(summary(forward.subset)$cp)) #Best cp Model
##  (Intercept)        Walks       CAtBat        CHits        CRuns         CRBI
##  110.1015291    5.2753942   -0.4498891    1.0510348    1.3425171    0.9381944
##       CWalks    DivisionW      PutOuts      Assists       Errors   NewLeagueN
##   -0.5426157 -125.5159872    0.1609710    0.4662319   -7.6994411   74.6774479
coef(forward.subset, which.max(summary(forward.subset)$adjr2)) #Best Adjusted R squ
ared Model
##  (Intercept)        AtBat         Hits        Walks        Years       CAtBat
##  207.8865937   -1.6603271    4.9552022    6.0893338  -12.4996617   -0.2504738
##        CHits        CRuns         CRBI       CWalks    DivisionW      PutOuts
##    0.3954684    1.4147636    1.0607231   -0.7225505 -123.4055931    0.1923965
##      Assists       Errors   NewLeagueN
##    0.5131110   -7.0115407   70.8774554
m.cp3=lm(Salary~Walks+CAtBat+CHits+CRuns+CRBI+CWalks+Division+PutOuts+Assists+Error
s+NewLeague,data=train.data)
m.bic3=lm(Salary~Walks+CAtBat+CHits+CRuns++CRBI+CWalks+Division,data=train.data)
m.adjr3=lm(Salary~AtBat+Hits+Walks+Years+CAtBat++CHits+CRuns+CRBI+CWalks+Division+P
utOuts+Assists+Errors+NewLeague,data=train.data)


mse.train.cp3=1/210*(sum((train.data$Salary-predict(m.cp3))^2)) #Training MSE for c
p
```

```
mse.train.bic3=1/210*(sum((train.data$Salary-predict(m.bic3))^2)) #Training MSE for
bic

mse.train.adjr3=1/210*(sum((train.data$Salary-predict(m.adjr3))^2)) #Training MSE f
or adjR2


y.hat.test.cp3=predict(m.cp3, newdata=test.data)

y.hat.test.bic3=predict(m.bic3, newdata=test.data)

y.hat.test.adjr3=predict(m.adjr3, newdata=test.data)

mse.test.cp3=mean((test.data$Salary-y.hat.test.cp3)^2)

mse.test.bic3=mean((test.data$Salary-y.hat.test.bic3)^2)

mse.test.adjr3=mean((test.data$Salary-y.hat.test.adjr3)^2)


out.forward=data.frame(Measure=c("Cp","BIC","AdjR2"),Training_MSE=c(mse.train.cp3,m
se.train.bic3,mse.train.adjr3),

                        Test_MSE=c(mse.test.cp3,mse.test.bic3,mse.test.adjr3))

out.forward

##   Measure Training_MSE Test_MSE

## 1      Cp     90888.44 145923.9

## 2     BIC     95658.81 146540.9

## 3   AdjR2     88287.08 134001.7

final=list(Best_Subset=out.best,Forward_Subset=out.forward,Backward_Subset=out.forw
ard)

final

## $Best_Subset

##   Measure Training_MSE Test_MSE

## 1      Cp     89790.63 125652.3

## 2     BIC     97026.43 153090.5

## 3   AdjR2     88935.80 130066.9

##

## $Forward_Subset

##   Measure Training_MSE Test_MSE

## 1      Cp     90888.44 145923.9

## 2     BIC     95658.81 146540.9

## 3   AdjR2     88287.08 134001.7

##

## $Backward_Subset

##   Measure Training_MSE Test_MSE

## 1      Cp     90888.44 145923.9

## 2     BIC     95658.81 146540.9

## 3   AdjR2     88287.08 134001.7

# 6. Backward Selection
```

```
backward.subset=regsubsets(Salary~.,data=train.data,nvmax=19,method="backward")
summary(backward.subset)
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = train.data, nvmax = 19,
##     method = "backward")
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: backward
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns CRBI
## 1  ( 1 )  " "   " "  " "   " "  " " " "  " "   " "    "*"   " "    " "   " "
## 2  ( 1 )  " "   " "  " "   " "  " " "*"  " "   " "    "*"   " "    " "   " "
## 3  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   " "    " "   " "
## 4  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 5  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 6  ( 1 )  " "   " "  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 7  ( 1 )  " "   "*"  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 8  ( 1 )  "*"   "*"  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 9  ( 1 )  "*"   "*"  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
## 10  ( 1 ) "*"   "*"  " "   " "  " " "*"  " "   "*"    "*"   "*"    " "   " "
```

```
## 11  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    "*"   "*"    "*"   " "
## 12  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    "*"   "*"    "*"   " "
## 13  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    "*"   "*"    "*"   " "
## 14  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   "*"    "*"   " "
## 15  ( 1 ) "*"   "*"  " "   " "  "*" "*"   "*"   "*"    "*"   "*"    "*"   " "
## 16  ( 1 ) "*"   "*"  "*"   " "  "*" "*"   "*"   "*"    "*"   "*"    "*"   " "
## 17  ( 1 ) "*"   "*"  "*"   " "  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 18  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 19  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
##           CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 2  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 3  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 4  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 5  ( 1 )  " "    " "     "*"       " "     " "     " "    " "
## 6  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 7  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 8  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 9  ( 1 )  " "    " "     "*"       "*"     "*"     " "    " "
## 10  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    " "
## 11  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    " "
## 12  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    "*"
## 13  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 14  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 15  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 16  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 17  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 18  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 19  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
coef(backward.subset, which.min(summary(backward.subset)$bic)) #Best BIC Model
##  (Intercept)        Walks       CAtBat        CHits       CHmRun    DivisionW
##  149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -129.0165012
coef(backward.subset, which.min(summary(backward.subset)$cp)) #Best cp Model
##  (Intercept)        AtBat         Hits        Walks       CAtBat        CHits
##  151.3709120   -1.5393497    4.6939494    6.1305798   -0.3519984    1.1424820
##       CHmRun        CRuns       CWalks    DivisionW      PutOuts      Assists
##    2.3082206    0.9269527   -0.5968973 -120.5643234    0.1975004    0.5618678
##       Errors   NewLeagueN
##   -7.1940771   71.8060203
```

```
coef(backward.subset, which.max(summary(backward.subset)$adjr2)) #Best Adjusted R s
quared Model

##  (Intercept)        AtBat         Hits        Walks       CAtBat        CHits

##  151.3709120   -1.5393497    4.6939494    6.1305798   -0.3519984    1.1424820

##       CHmRun        CRuns       CWalks    DivisionW      PutOuts      Assists

##    2.3082206    0.9269527   -0.5968973 -120.5643234    0.1975004    0.5618678

##       Errors   NewLeagueN

##   -7.1940771   71.8060203

m.cp2=lm(Salary~AtBat+Hits+Walks+CAtBat+CHits+CHmRun+CRuns+CWalks+Division+PutOuts+
Assists+Errors+NewLeague,data=train.data)

m.bic2=lm(Salary~Walks+CAtBat+CHits+CHmRun+Division,data=train.data)

m.adjr2=lm(Salary~AtBat+Hits+Walks+CAtBat+CHits+CHmRun+CRuns+CWalks+Division+PutOut
s+Assists+Errors+NewLeague,data=train.data)


mse.train.cp2=1/210*(sum((train.data$Salary-predict(m.cp2))^2)) #Training MSE for c
p

mse.train.bic2=1/210*(sum((train.data$Salary-predict(m.bic2))^2)) #Training MSE for
bic

mse.train.adjr2=1/210*(sum((train.data$Salary-predict(m.adjr2))^2)) #Training MSE f
or adjR2


y.hat.test.cp2=predict(m.cp2, newdata=test.data)

y.hat.test.bic2=predict(m.bic2, newdata=test.data)

y.hat.test.adjr2=predict(m.adjr2, newdata=test.data)

mse.test.cp2=mean((test.data$Salary-y.hat.test.cp2)^2)

mse.test.bic2=mean((test.data$Salary-y.hat.test.bic2)^2)

mse.test.adjr2=mean((test.data$Salary-y.hat.test.adjr2)^2)


out.backward=data.frame(Measure=c("Cp","BIC","AdjR2"),Training_MSE=c(mse.train.cp2,
mse.train.bic2,mse.train.adjr2),

                    Test_MSE=c(mse.test.cp2,mse.test.bic2,mse.test.adjr2))

out.backward

##   Measure Training_MSE Test_MSE

## 1      Cp     88865.02 134736.7

## 2     BIC     97026.43 153090.5

## 3   AdjR2     88865.02 134736.7
```

7. Comparison of Best Subset, Forward and Backward Selection
8. In terms of Model Selection

Best Subset Selection examines all possible models and gives the most optimal model for each criterion (Adj R^2, BIC, Cp) as :

Forward Selection :

1. Starts with no variables and adds step-by-step.

2. May miss the best combination of predictors.

Backward Selection :

1.Starts with full model and removes variables.

2.Works well but depends on initial full model.

2. In terms of Training MSE:

Best Subset → Lowest Training MSE (best fit)

Forward & Backward → Slightly higher Training MSE

3. In terms of Test MSE:

BIC-based models (often from any method) → Lower Test MSE

Best Subset (Adj R^2) → May have higher Test MSE (overfitting)

Forward/Backward → Comparable but sometimes slightly worse